

Cellulose Aerogel Simulation Literature Review

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User prompt: I want to perform computer simulations to predict mechanical properties of cellulose aerogels. Please help me with a initial bibliographical review (pass me a list of academic references) of studies that have made such simulations, to learn about the topic.

Response: Simulating the mechanical behavior of cellulose aerogels spans multiple length scales—from nanometer-scale cellulose nanofibrils (CNFs) to millimeter-scale porous networks.

1. Multi-Scale Simulation Approaches

Modeling Scale	Typical Method	Physical Focus	Key Mechanical Outputs
Atomistic (~1–10 nm)	All-Atom Molecular Dynamics (AAMD)	Inter-chain hydrogen bonding, crystal slip, chain cleavage	Intrinsic Young's modulus (E), ultimate tensile strength, inter-chain shear
Mesoscale (~10–500 nm)	Coarse-Grained MD (CG-MD) / Discrete Element Method (DEM)	CNF entanglements, nanofiber bending, cross-linking junction dynamics	Network flexibility, fiber entanglements, strain-induced reorientation
Continuum / Macro (>1 μ m)	Finite Element Method (FEM) / Voronoi Tessellations	Cellular network buckling, pore collapse under compression, macro-stress distributions	Bulk compressive modulus, Poisson's ratio, stress-strain response up to 80–90% strain

2. Recommended Literature & Foundational Studies

Category A: Atomistic & Molecular Dynamics Studies

Focus: Predicting the intrinsic mechanical limits of cellulose crystalline and amorphous domains, as well as fiber-fiber interfacial adhesion.

1. Elf, P., Özeren, H. D., Larsson, P. A., Larsson, A., Wågberg, L., Nilsson, R., Chaipayutham, P. T., Hedenqvist, M. S., & Nilsson, F. (2023). *Molecular Dynamics Simulations of Cellulose and Dialcohol Cellulose under Dry and Moist Conditions*. *Biomacromolecules*, 24(6), 2706–2720. <https://doi.org/10.1021/acs.biomac.3c00156>

- **Key Focus:** Explores atomistic tensile properties, hydrogen-bond dynamics, and structural mobility in modified cellulosic chains under dry and hydrated environments.

2. Khazraji, A. C., & Robert, S. (2013). *Interaction effects and mechanical properties of cellulose I β nanoparticles*. *Journal of Nanomaterials*, 2013.

- **Key Focus:** All-Atom MD simulation analyzing force fields (CHARMM/GROMOS) to obtain axial Young's modulus and elastic constants of cellulose nanocrystals (CNCs/CNFs).

Category B: Mesoscale & Coarse-Grained Models

Focus: Simulating assembly, fiber network deformation, and gelation mechanisms.

3. **Jarms, J. et al. (2024).** *Modeling of the Gelation Process and Network Architecture in Cellulose Aerogels.* **ACS Applied Bio Materials / Biomacromolecules.**

- **Key Focus:** Uses Discrete Element Modeling (DEM) and Langevin dynamics to simulate how cellulose chains assemble into porous hydrogels and aerogels, providing a structural framework for subsequent mechanical testing.

4. **Sinko, R., & Keten, S. (2014).** *Traction-separation laws for hydrogen-bonded plant cell wall interfaces.* **Applied Physics Letters**, 105(24), 243702.

- **Key Focus:** Coarse-grained modeling of cellulose nanofibril interfaces under shearing and peeling modes, providing interfacial force laws required for larger-scale network simulations.

Category C: Microstructural Mechanics & Finite Element Modeling (FEM)

Focus: Simulating cellular collapse, porous geometry, and macro-scale compression curves.

5. **Rege, A., Preibisch, P., & Gurlo, A. (2021).** *Modeling the structure–property relationships in biopolymer aerogels.* **Gels**, 7(3), 112.

- **Key Focus:** Constitutive mechanical models and beam-based finite element networks designed to predict the nonlinear compressive stress-strain response (elastic, plateau, and densification regimes) of bio-aerogels.

6. **Chandrasekaran, S. et al. (2018).** *Microstructural modeling of cellular open-porous biopolymer aerogels using Laguerre–Voronoi tessellation.* **Materials & Design**, 155, 370–380.

- **Key Focus:** Uses random closed packing of polydisperse spheres and Laguerre–Voronoi tessellations to construct realistic 3D open-cell FEM geometries without fitting parameters.

7. **Paulauskiene, T., Sirtaute, E., Tadzijevas, A., & Uebe, J. (2024).** *Mechanical Properties of Cellulose Aerogel Composites with and without Crude Oil Filling.* **Gels**, 10(2), 135.
<https://doi.org/10.3390/gels10020135>

- **Key Focus:** Experimental compression benchmarks (yielding, elastic recovery, and plasticization effects) useful for validating numerical constitutive models.

3. Recommended Simulation Tools & Software

- **Molecular Dynamics (Atomistic/CG):** LAMMPS or GROMACS (using CHARMM36 or REAXFF force fields for cellulose).

- **Network Generation & Voronoi Meshing: Neper or Rhino/Grasshopper** (for generating Laguerre-Voronoi cellular microstructures).
 - **Finite Element Analysis: ABAQUS or ANSYS** (using Timoshenko beam elements or 3D solid elements for cellular wall crushing).
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